F70LA – Life Insurance Mathematics A, 2025/26
Chapter 5 – Life annuities

A **life annuity** is a series of payments that continue as long as the recipient is alive. The recipient is called the **annuitant**.

The last chapter defined various different functions for the expected present value of benefits paid to the policyholder. In this chapter, we consider the other side of the equation, that is the expected present value of premiums paid to the policyholder. These can be thought of as life annuities, since the policyholder will only pay premiums as long as they are alive.

Annuities are also common pension benefits, paid during the retirement of a member of a pension scheme.

5.1 Review of annuities-certain

Fix an integer $n > 0$ representing the number of years for which a payment is made, i.e. the term. You should already have seen the following annuities-certain, or fixed-term annuities, in a previous course.

- **Annuity-certain payable in advance.** $\ddot{a}_{\overline{n}|}$ represents the present value of an annuity paying £1 per annum annually in advance (i.e. at the start of each year) for n years, with
$$\ddot{a}_{\overline{n}|} = 1 + v + v^2 + \cdots + v^{n-1} = \frac{1 - v^n}{d}.$$
This annuity is also called an annuity in advance.

- **Annuity-certain payable in arrears** $a_{\overline{n}|}$ represents the present value of an annuity paying £1 per annum annually in arrears (i.e. at the end of each year) for n years, with
$$a_{\overline{n}|} = v + v^2 + \cdots + v^n = \frac{1 - v^n}{i}.$$

- **Annuity-certain payable continuously** $\bar{a}_{\overline{n}|}$ represents the present value of an annuity payable continuously for n years at the rate £1 per annum, with
$$\bar{a}_{\overline{n}|} = \int_0^n v^t \, dt = \frac{1 - v^n}{\delta}.$$

- m**thly-paid annuities-certain** When the payment of £1 is broken up into an integer $m \geq 1$ payments of amount £$\frac{1}{m}$ made every $1/m$th year, the present values of the relevant annuities are
$$\ddot{a}^{(m)}_{\overline{n}|} = \frac{1}{m} \cdot 1 + \frac{1}{m} \cdot v^{\frac{1}{m}} + \frac{1}{m} \cdot v^{\frac{2}{m}} + \cdots + \frac{1}{m} \cdot v^{\frac{nm-1}{m}} = \frac{1 - v^n}{d^{(m)}}$$
and
$$a^{(m)}_{\overline{n}|} = \frac{1}{m} \cdot v^{\frac{1}{m}} + \frac{1}{m} \cdot v^{\frac{2}{m}} + \cdots + \frac{1}{m} \cdot v^{\frac{nm}{m}} = \frac{1 - v^n}{i^{(m)}}.$$

- **Arithmetically increasing annuities-certain** If the payment is £1 in the first year, £2 in the second year, £3 in the third year, and so on, the present values of these increasing payments paid in advance and in arrears for n years are

$$(I\ddot{a})_x = \frac{\ddot{a}_{\overline{n}|} - n\,v^n}{d} \qquad \text{and} \qquad (Ia)_x = \frac{a_{\overline{n}|} - n\,v^n}{i},$$

respectively.

Handy hint 5.1 *You can find the formulae* $(Ia)_x = \frac{a_{\overline{n}|} - n\,v^n}{i}$ *in* Formulae and Tables for Actuarial Examinations*, in the* Formulae *section, Section 4. However, you are more likely to need* $(I\ddot{a})_x$*, so make sure you either learn its formula (replace* i *by* d *in the the formulae for* $(Ia)_x$*).*

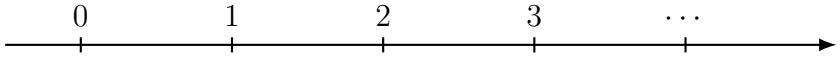
Read Dickson, Hardy & Waters, Chapter 5.

5.2 Whole life annuity-due (i.e payments in advance)

Suppose that:

- Payments of £1 are made at the start of each year,...
- ...As long as a life, initially aged x years, is alive.

This is a life annuity: a series of payments made as long as the annuitant is alive. With the timing of payments at the start of each year, it is called a *whole life annuity-due*.



Amount	£1	£1	£1	£1	···
Discount	1	v	v^2	v^3	···
Probability	1	p_x	${}_2p_x$	${}_3p_x$	···

Let the effective rate of interest be $i > -1$ p.a. At time $k \in \{0, 1, 2, \ldots\}$, a payment is made as long as (x) is alive. In random variable terms, this means $K_x \geq k$. By using the indicator random variable

$$\mathbb{1}_{[K_x \geq k]} := \begin{cases} 1 & \text{if } K_x \geq k, \\ 0 & \text{otherwise,} \end{cases}$$

we can indicate if an annuity payment should be made at time k (i.e. $\mathbb{1}_{[K_x \geq k]} = 1$, corresponding to (x) alive at time k) or not (i.e. $\mathbb{1}_{[K_x \geq k]} = 0$, corresponding to (x) dying before time k).

The present value random variable of these payments can thus be defined as

$$Y := \sum_{k=0}^{\infty} \mathbb{1}_{[K_x \geq k]}\, v^k.$$

The expected value of Y is $\ddot{a}_x := \mathbb{E}(Y)$. As K_x takes only positive integer values,

$$\mathbb{E}(Y) = \mathbb{E}\left(\sum_{k=0}^{\infty} \mathbb{1}_{[K_x \geq k]}\, v^k\right) = \sum_{k=0}^{\infty} \mathbb{P}[K_x \geq k]\, v^k.$$

As $[K_x \geq k]$ is the event that x survives at least k years, then $\mathbb{P}[K_x \geq k] = {}_kp_x$ and so

$$\boxed{\ddot{a}_x = \sum_{k=0}^{\infty} v^k\, {}_kp_x.}$$

Exercise 5.2 *Calculate the EPV of a whole life annuity of £500 per annum payable in advance to* (60)*, if* $\ddot{a}_{60} = 11.891$ *for a given survival model and an annual effective interest rate of* 0.06*.*

The EPV is $500\,\ddot{a}_{60} = 500 \times 11.891 = £5,945.50$.

Example 5.3 *Show that*

$$\boxed{\ddot{a}_x = \frac{1 - A_x}{d}.}$$

First, noting that there are $K_x + 1$ *terms in the sum* $\sum_{k=0}^{K_x}$*,*

$$Y = \sum_{k=0}^{\infty} \mathbb{1}_{[K_x \geq k]}\, v^k = \sum_{k=0}^{K_x} v^k = \ddot{a}_{\overline{K_x+1}|} = \frac{1 - v^{K_x+1}}{d}.$$

Thus

$$\ddot{a}_x = \mathbb{E}(Y) = \mathbb{E}\left(\frac{1 - v^{K_x+1}}{d}\right) = \frac{1 - \mathbb{E}(v^{K_x+1})}{d}.$$

Since v^{K_x+1} *represents a payment of 1 unit paid at the end of year of death of* (x)*, then* $\mathbb{E}(v^{K_x+1}) = A_x$*. Substituting into the last fraction, we get the result.*

For any random variable X and constants a, b, $\mathbb{V}ar(aX + b) = a^2\mathbb{V}ar(X)$. Also recall that $\mathbb{V}ar\left(v^{K_x+1}\right) = {}^2A_x - (A_x)^2$. Hence

$$\mathbb{V}ar(\ddot{a}_{\overline{K_x+1}|}) = \mathbb{V}ar\left(\frac{1 - v^{K_x+1}}{d}\right) = \frac{1}{d^2}\mathbb{V}ar\left(v^{K_x+1}\right) = \frac{{}^2A_x - (A_x)^2}{d^2}.$$

Exercise 5.4 *Calculate the EPV of a whole life annuity of £500 per annum payable in advance to* (60)*, if* $A_{60} = 0.32692$ *for a given survival model and an interest rate of 0.06 per annum effective.*

The EPV is

$$500\,\ddot{a}_{60} = 500 \times \left(\frac{1 - A_{60}}{d}\right) = 500 \times \left(\frac{1 - 0.32692}{1 - 1.06^{-1}}\right) = 500 \times 11.89108 = £5,945.54.$$

Exercise 5.5 *Show that the variance of a whole life annuity payable in advance to* (x)*, valued at an effective rate of interest* i *p.a, is given by*

$$\mathbb{V}ar(Y) = \mathbb{V}ar(\ddot{a}_{\overline{K_x+1|}}) = \frac{{}^2A_x - (A_x)^2}{d^2}.$$

For any random variable X and constants a, b, $\mathbb{V}ar(aX + b) = a^2\mathbb{V}ar(X)$. Also recall that $\mathbb{V}ar\left(v^{K_x+1}\right) = {}^2A_x - (A_x)^2$. Hence

$$\mathbb{V}ar(\ddot{a}_{\overline{K_x+1|}}) = \mathbb{V}ar\left(\frac{1 - v^{K_x+1}}{d}\right) = \frac{1}{d^2}\mathbb{V}ar\left(v^{K_x+1}\right) = \frac{{}^2A_x - (A_x)^2}{d^2}.$$

Exercise 5.6 *Calculate the standard deviation of the present value of a whole life annuity of £500 per annum payable in advance to* (60)*, if* $A_{60} = 0.32692$ *and* ${}^2A_{60} = 0.14098$ *for a given survival model and an an interest rate of 0.06 per annum effective.*

As the present value random variable is $500\ddot{a}_{\overline{K_{60}+1|}}$, the variance is

$$\mathbb{V}ar(500\ddot{a}_{\overline{K_{60}+1|}}) = (500)^2\,\mathbb{V}ar(\ddot{a}_{\overline{K_{60}+1|}}) = (500)^2\,\frac{{}^2A_{60x} - (A_{60})^2}{d^2}.$$

Thus the standard deviation is

$$(500)\,\frac{({}^2A_{60} - (A_{60})^2)^{0.5}}{d} = (500)\,\frac{(0.14098 - (0.32692)^2)^{0.5}}{1 - 1.06^{-1}} = 500 \times 3.26252 = £1,631.26.$$

Handy hint 5.7 *You can find the formulae*

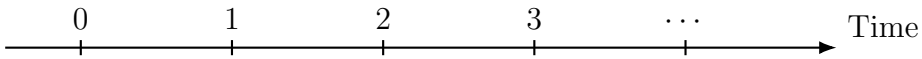
$$\ddot{a}_x = \frac{1 - A_x}{d}, \quad \mathbb{V}ar(\ddot{a}_{\overline{K_x+1|}}) = \frac{{}^2A_x - (A_x)^2}{d^2} \quad \textit{and} \quad \ddot{a}_x = \mathbb{E}\left(\ddot{a}_{\overline{K_x+1|}}\right)$$

in Formulae and Tables for Actuarial Examinations*, in the* Formulae *section, Section 6.*

5.3 Whole life immediate annuity (payable in arrears)

To value a *whole life immediate annuity*, which means the payments are at the end of each time period, consider an annuity where:

- Payments of £1 are made at the end of each year,...
- ...As long as a life, initially aged x years, is alive.



Amount	£1	£1	£1	· · ·
Discount	v	v^2	v^3	· · ·
Probability	p_x	${}_2p_x$	${}_3p_x$	· · ·

Let the effective rate of interest be $i > -1$ p.a. The present value random variable of these payments is (noting the indexation starts at $k = 1$)

$$Y^\star := \sum_{k=1}^{\infty} \mathbb{1}_{[K_x \geq k]}\, v^k = Y - 1,$$

where the random variable Y is the present value random variable of the comparable annuity-due, defined in Section 5.2.

The expected value of $Y^\star$ is $a_x := \mathbb{E}\,(Y^\star)$. As $\mathbb{E}\,(Y) = \ddot{a}_x$,

$$\boxed{a_x = \ddot{a}_x - 1 = \sum_{k=1}^{\infty} v^k\, {}_kp_x.}$$

Exercise 5.8 *Show that the variance of a whole life immediate annuity to (x), valued at an effective rate of interest i p.a, is given by*

$$\mathbb{V}ar(Y^\star) = \frac{{}^2A_x - (A_x)^2}{d^2}.$$

For any random variable X and constants a, b, $\mathbb{V}ar(aX + b) = a^2\mathbb{V}ar(X)$. Hence using the result of Exercise 5.5,

$$\mathbb{V}ar(Y^\star) = \mathbb{V}ar(Y - 1) = \mathbb{V}ar\,(Y) = \frac{{}^2A_x - (A_x)^2}{d^2}.$$

5.4 Whole life continuous annuity

To value a *whole life continuous annuity*, consider

- Payments made continuously at the rate of £1 per annum,...
- ...As long as a life, initially aged x years, is alive.

Thus the payments continue as long as (x) is alive. The event that (x) is alive at time $t > 0$ is $[T_x > t]$. By using the indicator random variable

$$\mathbb{1}_{[T_x>t]} := \begin{cases} 1 & \text{if } T_x > t, \\ 0 & \text{otherwise,} \end{cases}$$

we can indicate whether the payments continue at time t (i.e. states of the world ω in which $\mathbb{1}_{[T_x>t]}(\omega) = 1$, meaning (x) is still alive at time t) or have stopped (i.e. states of the world ω in which $\mathbb{1}_{[T_x>t]}(\omega) = 0$, meaning (x) died before time t).

Let the force of interest be δ p.a. The present value random variable Y of the whole life continuous annuity can thus be written as

$$Y = \int_0^\infty e^{-\delta t}\, \mathbb{1}_{[T_x>t]}\, dt.$$

The expected value of Y is $\bar{a}_x := \mathbb{E}(Y)$. As $\mathbb{E}(\mathbb{1}_{[T_x>t]}) = \mathbb{P}[T_x > t] = {}_tp_x$,

$$\mathbb{E}(Y) = \mathbb{E}\left(\int_0^\infty e^{-\delta t}\, \mathbb{1}_{[T_x>t]}\, dt\right) = \int_0^\infty e^{-\delta t}\, \mathbb{E}(\mathbb{1}_{[T_x>t]})\, dt = \int_0^\infty e^{-\delta t}\, {}_tp_x\, dt,$$

and we get

$$\boxed{\bar{a}_x = \int_0^\infty e^{-\delta t}\, {}_tp_x\, dt.}$$

The expression for the EPV $\bar{a}_x$ can be interpreted as follows:

$$\bar{a}_x = \underbrace{\int_0^\infty}_{\text{Payment times}} \underbrace{e^{-\delta t}}_{\text{Discount}}\ \underbrace{{}_tp_x}_{\text{Probability}}\ \underbrace{1 \cdot dt}_{\text{Instantaneous amount paid}}.$$

Remark 5.9 *There are various ways of deriving the expression $\bar{a}_x = \int_0^\infty e^{-\delta t}\, {}_tp_x\, dt$, which you can read in Dickson, Hardy & Waters, Section 5.5.1.*

Example 5.10 *Show that*

$$\boxed{\bar{a}_x = \frac{1 - \bar{A}_x}{\delta}.}$$

From

$$Y = \int_0^\infty e^{-\delta t}\, \mathbb{1}_{[T_x>t]}\, dt = \int_0^{T_x} e^{-\delta t}\, dt = \bar{a}_{\overline{T_x}|},$$

substitute for $\bar{a}_{\overline{T_x}|} = (1 - v^{T_x})/\delta$ *and take expectations to get*

$$\bar{a}_x = \mathbb{E}(Y) = \mathbb{E}\left(\frac{1 - v^{T_x}}{\delta}\right) = \frac{1 - \mathbb{E}\left(v^{T_x}\right)}{\delta} = \frac{1 - \bar{A}_x}{\delta}.$$

Exercise 5.11 *For a whole life annuity of £1 per annum payable continuously to a person aged* (x)*, show that the variance of the present value random variable* Y*, at a force of interest* δ *per annum, is*

$$\mathbb{V}ar(Y) = \frac{{}^2\bar{A}_x - \left(\bar{A}_x\right)^2}{\delta^2}.$$

For this exercise, it is easier to start with

$$Y = \bar{a}_{\overline{T_x}|}.$$

Recall that $\mathbb{V}ar\left(v^{T_x}\right) = {}^2\bar{A}_x - (\bar{A}_x)^2$. Hence

$$\mathbb{V}ar(Y) = \mathbb{V}ar(\bar{a}_{\overline{T_x}|}) = \mathbb{V}ar\left(\frac{1 - v^{T_x}}{\delta}\right) = \frac{1}{\delta^2}\mathbb{V}ar\left(v^{T_x}\right) = \frac{{}^2\bar{A}_x - (\bar{A}_x)^2}{\delta^2}.$$

Exercise 5.12 *Explain why for any survival model and evaluated at any effective rate of interest* $i > -1$*,*

$$a_x < \bar{a}_x < \ddot{a}_x.$$

Read Dickson, Hardy and Waters, Section 5.7 for an explanation and/or to check your answer.

Handy hint 5.13 *You can find the formulae*

$$\bar{a}_x = \frac{1 - \bar{A}_x}{\delta}, \quad \mathbb{V}ar(\bar{a}_{\overline{T_x}|}) = \frac{{}^2\bar{A}_x - \left(\bar{A}_x\right)^2}{\delta^2} \quad \textit{and} \quad \bar{a}_x = \mathbb{E}\left(\bar{a}_{\overline{T_x}|}\right)$$

in Formulae and Tables for Actuarial Examinations*, in the* Formulae *section, Section 6.*

5.5 Term immediate annuity (i.e payments in arrears)

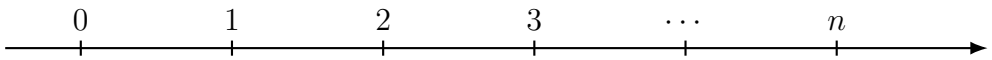
Often, premiums are only paid for a limited time as long as the policyholder is alive. This is true for endowment insurances, where the premiums are paid at most until the survival benefit is paid. Premiums stop once the policyholder dies. For this reason, we value these payment streams which are called *term annuities*.

To value a *term immediate annuity*, consider an annuity where:

- Payments of £1 are made at the end of each year,...
- ...As long as a life, initially aged x years, is alive and

- ...For a maximum of an integer $n > 0$ years.

This means there are at most n payments made.



Amount	£1	£1	£1	⋯	£1
Discount	v	v^2	v^3	⋯	v^n
Probability	p_x	${}_2p_x$	${}_3p_x$	⋯	${}_np_x$

Let the effective rate of interest be $i > -1$ p.a. The present value random variable of these payments is

$$Y := \sum_{k=1}^{n} \mathbb{1}_{[K_x \geq k]}\, v^k.$$

The expected value of Y is $a_{x:\overline{n}|} := \mathbb{E}(Y)$. As K_x takes only positive integer values,

$$\mathbb{E}(Y) = \mathbb{E}\left(\sum_{k=1}^{n} \mathbb{1}_{[K_x \geq k]}\, v^k\right) = \sum_{k=1}^{n} \mathbb{P}[K_x \geq k]\, v^k.$$

As $[K_x \geq k]$ is the event that x survives at least k years, then $\mathbb{P}[K_x \geq k] = {}_kp_x$ and so

$$\boxed{a_{x:\overline{n}|} = \sum_{k=1}^{n} v^k\, {}_kp_x.}$$

Exercise 5.14 *Explain why*

$$Y = a_{\overline{\min\{K_x,n\}}|}.$$

If in a future state of the world ω, it happens that $K_x(\omega) < n$, then exactly $K_x(\omega)$ payments will have been made, at times $1, 2, \ldots, K_x(\omega)$. If in another future state of the world $\tilde{\omega}$, it happens that $K_x(\tilde{\omega}) \geq n$, meaning (x) survives for at least n years, then exactly n payments will have been made, since the payments stop at time n.

In summary, the number of payments is $\min\{K_x, n\}$. The present value of $\min\{K_x, n\}$ annuity payments of £1 paid at the end of each year is

$$Y = a_{\overline{\min\{K_x,n\}}|}.$$

Exercise 5.15 *Show that the variance of Y, the present value of a n-year temporary immediate annuity which is valued at an effective rate of interest $i > -1$ p.a, is given by*

$$\mathbb{V}ar(Y) = \frac{{}^2A_{x:\overline{n+1}|} - (A_{x:\overline{n+1}|})^2}{d^2}.$$

Using the expression

$$Y = a_{\overline{\min\{K_x,n\}|}} = \frac{1 - v^{\min\{K_x,n\}}}{i},$$

we get

$$\mathbb{V}ar(Y) = \mathbb{V}ar\left(\frac{1 - v^{\min\{K_x,n\}}}{i}\right) = \frac{1}{i^2}\,\mathbb{V}ar\left(v^{\min\{K_x,n\}}\right).$$

Recall that for an integer $m > 0$, the random variable

$$Z := v^{\min\{K_x+1,m\}} \qquad \Rightarrow \qquad \mathbb{E}\left(Z\right) = A_{x:\overline{m|}} \quad \text{and} \quad \mathbb{V}ar\left(Z\right) = {}^2A_{x:\overline{m|}} - \left(A_{x:\overline{m|}}\right)^2.$$

Then recalling that $v^{-1} = 1 + i$ and $d = \frac{i}{1+i}$,

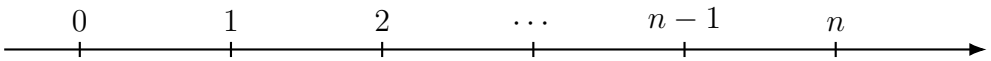
$$\begin{aligned}\mathbb{V}ar(Y) = \frac{1}{i^2}\,\mathbb{V}ar\left(v^{\min\{K_x,n\}}\right) = \frac{1}{i^2}\,\mathbb{V}ar\left(v^{\min\{K_x+1,n+1\}-1}\right) &= \left(\frac{1+i}{i}\right)^2 \mathbb{V}ar\left(v^{\min\{K_x+1,n+1\}}\right)\\ &= \frac{1}{d^2}\,\mathbb{V}ar\left(v^{\min\{K_x+1,n+1\}}\right)\\ &= \frac{{}^2A_{x:\overline{n+1|}} - A_{x:\overline{n+1|}}}{d^2}.\end{aligned}$$

5.6 Term annuity-due (i.e payments in advance)

To value a *term annuity-due*, consider an annuity where:

- Payments of £1 are made at the start of each year,...
- ...As long as a life, initially aged x years, is alive and
- ...For a maximum of an integer $n > 0$ years.

This means there are at most n payments made.



Amount	£1	£1	£1	⋯	£1
Discount	1	v	v^2	⋯	v^{n-1}
Probability	1	p_x	${}_2p_x$	⋯	${}_{n-1}p_x$

Let the effective rate of interest be $i > -1$ p.a. The present value random variable of these payments is

$$Y := \sum_{k=0}^{n-1} \mathbb{1}_{[K_x \geq k]}\, v^k.$$

The expected value of Y is

$$\ddot{a}_{x:\overline{n}|} := \mathbb{E}(Y) = \mathbb{E}\left(\sum_{k=0}^{n-1} \mathbb{1}_{[K_x \geq k]}\, v^k\right) = \sum_{k=0}^{n-1} \mathbb{P}[K_x \geq k]\, v^k = \sum_{k=0}^{n-1} v^k\, {}_kp_x,$$

i.e.

$$\boxed{\ddot{a}_{x:\overline{n}|} = \sum_{k=0}^{n-1} v^k\, {}_kp_x.}$$

Exercise 5.16 *Explain why*

$$Y = \ddot{a}_{\overline{\min\{K_x+1,n\}}|}.$$

If in a future state of the world ω, it happens that $K_x(\omega) < n-1$, then exactly $K_x(\omega)+1$ payments will have been made, at times $0, 1, 2, \ldots, K_x(\omega)$. If in another future state of the world $\tilde{\omega}$, it happens that $K_x(\tilde{\omega}) \geq n-1$, meaning (x) survives for at least $n-1$ years, then exactly n payments will have been made, since the payments stop at time $n-1$.

In summary, the number of payments is $\min\{K_x+1, n\}$. The present value of $\min\{K_x+1, n\}$ annuity payments of £1 paid at the end of each year is

$$Y = \ddot{a}_{\overline{\min\{K_x+1,n\}}|}.$$

Exercise 5.17 *Show that the variance of Y, the present value of a n-year temporary annuity-due which is valued at an effective rate of interest $i > -1$ p.a, is given by*

$$\mathbb{V}ar(Y) = \frac{{}^2A_{x:\overline{n}|} - (A_{x:\overline{n}|})^2}{d^2}.$$

Using the expression

$$Y = \ddot{a}_{\overline{\min\{K_x+1,n\}}|} = \frac{1 - v^{\min\{K_x+1,n\}}}{d},$$

we get

$$\mathbb{V}ar(Y) = \mathbb{V}ar\left(\frac{1 - v^{\min\{K_x+1,n\}}}{d}\right) = \frac{1}{d^2}\,\mathbb{V}ar\left(v^{\min\{K_x+1,n\}}\right).$$

Recall that for an integer $m > 0$, the random variable

$$Z := v^{\min\{K_x+1,m\}} \qquad \Rightarrow \qquad \mathbb{E}(Z) = A_{x:\overline{m}|} \quad \text{and} \quad \mathbb{V}ar(Z) = {}^2A_{x:\overline{m}|} - \left(A_{x:\overline{m}|}\right)^2.$$

Then

$$\mathbb{V}ar(Y) = \frac{1}{d^2}\,\mathbb{V}ar\left(v^{\min\{K_x+1,n\}}\right) = \frac{{}^2A_{x:\overline{n}|} - A_{x:\overline{n}|}}{d^2}.$$

Exercise 5.18 *Show that*

$$\boxed{a_{x:\overline{n}|} = \ddot{a}_{x:\overline{n}|} - 1 + {}_{n}p_x.}$$

$$a_{x:\overline{n}|} = \sum_{k=1}^{n} v^k \, {}_{k}p_x = \sum_{k=0}^{n-1} v^k \, {}_{k}p_x - 1 + v^n \, {}_{n}p_x = \ddot{a}_{x:\overline{n}|} - 1 + {}_{n}p_x.$$

Exercise 5.19 *Show that*

$$\boxed{\ddot{a}_{x:\overline{n}|} = \ddot{a}_x - v^n \, {}_{n}p_x \, \ddot{a}_{x+n}.}$$

$$\begin{aligned}\ddot{a}_{x:\overline{n}|} = \sum_{k=0}^{n-1} v^k \, {}_{k}p_x = \sum_{k=0}^{\infty} v^k \, {}_{k}p_x - \sum_{k=n}^{\infty} v^k \, {}_{k}p_x = \ddot{a}_x - \sum_{r=0}^{\infty} v^{r+n} \, {}_{r+n}p_x &= \ddot{a}_x - v^n \, {}_{n}p_x \sum_{r=0}^{\infty} v^r \, {}_{r}p_{x+n} \\ &= \ddot{a}_x - v^n \, {}_{n}p_x \, \ddot{a}_{x+n}.\end{aligned}$$

5.7 Term continuous annuity

To value a *term continuous annuity*, consider

- Payments made continuously at the rate of £1 per annum,...
- ...As long as a life, initially aged x years, is alive and...
- ...For a maximum of an integer $n > 0$ years.

Let the force of interest be δ p.a. The present value random variable of these payments is

$$Y := \int_0^n e^{-\delta t} \, \mathbb{1}_{[T_x > t]} \, dt.$$

As $\mathbb{E}\left(\mathbb{1}_{[T_x>t]}\right) = \mathbb{P}[T_x > t] = {}_{t}p_x$, the expected value of Y is

$$\bar{a}_{x:\overline{n}|} := \mathbb{E}\left(Y\right) = \mathbb{E}\left(\int_0^n e^{-\delta t} \, \mathbb{1}_{[T_x>t]} \, dt\right) = \int_0^n e^{-\delta t} \, \mathbb{E}\left(\mathbb{1}_{[T_x>t]}\right) \, dt = \int_0^n e^{-\delta t} \, {}_{t}p_x \, dt,$$

i.e.

$$\boxed{\bar{a}_{x:\overline{n}|} = \int_0^n e^{-\delta t} \, {}_{t}p_x \, dt.}$$

Exercise 5.20 *Explain why*

$$Y = \bar{a}_{\overline{\min\{T_x, n\}}|}.$$

If in a future state of the world ω, it happens that $T_x(\omega) < n$, then payments last for exactly $T_x(\omega)$ years. If in another future state of the world $\tilde{\omega}$, it happens that $T_x(\tilde{\omega}) \geq n$, meaning (x) survives for at least n years, then exactly n years of payments will have been made, since the payments stop at time n.

In summary, the number of years of payments is $\min\{T_x, n\}$. The present value of $\min\{T_x, n\}$ years of annuity payments made at the continuous rate of £1 per annum is

$$Y = \bar{a}_{\overline{\min\{T_x,n\}|}}.$$

Exercise 5.21 *Show that the variance of Y, the present value of a n-year term continuous annuity which is valued at a force of interest δ p.a, is*

$$\mathbb{V}ar(Y) = \frac{{}^2\bar{A}_{x:\overline{n}|} - (\bar{A}_{x:\overline{n}|})^2}{\delta^2}.$$

Using the expression

$$Y = \bar{a}_{\overline{\min\{T_x,n\}|}} = \frac{1 - v^{\min\{T_x,n\}}}{\delta},$$

we get

$$\mathbb{V}ar(Y) = \mathbb{V}ar\left(\frac{1 - v^{\min\{T_x,n\}}}{\delta}\right) = \frac{1}{\delta^2}\,\mathbb{V}ar\left(v^{\min\{T_x,n\}}\right).$$

Recall that for an integer $m > 0$, the random variable

$$Z := v^{\min\{T_x,m\}} \qquad \Rightarrow \qquad \mathbb{E}\,(Z) = \bar{A}_{x:\overline{m}|} \quad \text{and} \quad \mathbb{V}ar\,(Z) = {}^2\bar{A}_{x:\overline{m}|} - \left(\bar{A}_{x:\overline{m}|}\right)^2.$$

Then

$$\mathbb{V}ar(Y) = \frac{1}{\delta^2}\,\mathbb{V}ar\left(v^{\min\{T_x,n\}}\right) = \frac{{}^2\bar{A}_{x:\overline{n}|} - \bar{A}_{x:\overline{n}|}}{\delta^2}.$$

Exercise 5.22 *Show that*

$$\boxed{\bar{a}_{x:\overline{n}|} = \bar{a}_x - v^n\,{}_np_x\,\bar{a}_{x+n}.}$$

Changing the variable of integration from t to $s = t - n$ at the end of the first line,

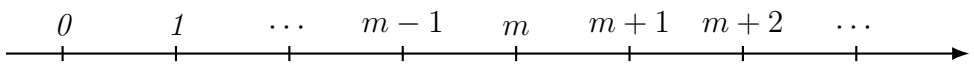
$$\begin{aligned}
\bar{a}_{x:\overline{n}|} = \int_0^n e^{-\delta t}\,{}_tp_x\,dt = \int_0^\infty e^{-\delta t}\,{}_tp_x\,dt - \int_n^\infty e^{-\delta t}\,{}_tp_x\,dt &= \bar{a}_x - \int_0^\infty e^{-\delta(s+n)}\,{}_{s+n}p_x\,dt \\
&= \ddot{a}_x - v^n\,{}_np_x \int_0^\infty e^{-\delta s}\,{}_sp_{x+n}\,dt \\
&= \bar{a}_x - v^n\,{}_np_x\,\bar{a}_{x+n}.
\end{aligned}$$

5.8 Deferred annuities

In an m-year deferred life annuity, the annuity payments begin in m years' time.

Example 5.23 *Consider an m-year deferred whole life annuity-due in which:*

- *Payments of £1 begin at time m, when (x) is age $(x+m)$ years,...*
- *With payments made at the start of each year from time m,...*
- *...As long as the life is alive.*



Amount		£1	£1	£1	...
Discount		v^m	v^{m+1}	v^{m+2}	...
Probability		${}_mp_x$	${}_{m+1}p_x$	${}_{m+2}p_x$	...

The expected present value of an m-year deferred whole life annuity-due is

$$ {}_{m|}\ddot{a}_x = \sum_{k=m}^{\infty} v^k \, {}_kp_x = \sum_{r=0}^{\infty} v^{r+m} \, {}_{r+m}p_x = v^m \, {}_mp_x \sum_{r=0}^{\infty} v^{r+m} \, {}_rp_{x+m} = v^m \, {}_mp_x \, \ddot{a}_{x+m}, $$

i.e.

$$ \boxed{ {}_{m|}\ddot{a}_x = v^m \, {}_mp_x \, \ddot{a}_{x+m}. } $$

Similar relationships hold for the other deferred annuities studied, e.g.

$$ \boxed{ {}_{m|}a_x = v^m \, {}_mp_x \, a_{x+m} \qquad \text{and} \qquad {}_{m|}\bar{a}_x = v^m \, {}_mp_x \, \bar{a}_{x+m}. } $$

Exercise 5.24 *Show that*

$$ \boxed{ {}_{m|}a_{x:\overline{n}|} = v^m \, {}_mp_x \, a_{x+m:\overline{n}|}. } $$

$$ {}_{m|}a_{x:\overline{n}|} = \sum_{k=m}^{m+n} v^k \, {}_kp_x = \sum_{r=0}^{n} v^{r+m} \, {}_{r+m}p_x = v^m \, {}_mp_x \sum_{r=0}^{n} v^{t+m} \, {}_rp_{x+m} = v^m \, {}_mp_x \, \ddot{a}_{x+m:\overline{n}|}. $$

Exercise 5.25 *Show that*

$$ {}_{m|}\ddot{a}_x = \ddot{a}_x - \ddot{a}_{x:\overline{m}|} $$

and explain the result.

Mathematically,

$$ {}_{m|}\ddot{a}_x = \sum_{k=m}^{\infty} v^k \, {}_kp_x = \sum_{k=0}^{\infty} v^k \, {}_kp_x - \sum_{k=0}^{m-1} v^k \, {}_kp_x = \ddot{a}_x - \ddot{a}_{x:\overline{m}|}. $$

The payments made under the whole life annuity-due, which have EPV $\ddot{a}_x$, for the first n years match exactly the payments made under the m-year term annuity-due, which have EPV $\ddot{a}_{x:\overline{m}|}$. Therefore, the net payments are £0 for the first m years.

The net payment is

- £1 at time m, if (x) survives to time m,
- £1 at time $m+1$, if (x) survives to time $m+1$,
- And so on.

The net payments match exactly those for an m-year deferred whole life annuity-due. Calculating their respective expected values, it must hold that ${}_{m|}\ddot{a}_x = \ddot{a}_x - \ddot{a}_{x:\overline{m}|}$.



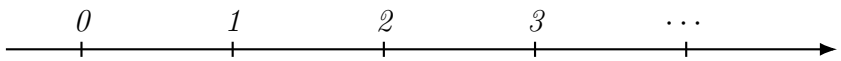
Cashflow for $\ddot{a}_x$	£1	£1	⋯	£1	£1	£1	⋯
Cashflow for $\ddot{a}_{x:\overline{m}|}$	£1	£1	⋯	£1			
Cashflow for $\ddot{a}_x - \ddot{a}_{x:\overline{m}|}$					£1	£1	⋯

5.9 Arithmetically increasing annuities

Annuity payments do not have to be the same each year. Here we consider annuity payments which increase arithmetically each year, which are called *arithmetically increasing annuities*.

Example 5.26 *Consider an arithmetically increasing whole life annuity-due in which:*

- *A payment of £1 is made in the first year, a payment of £2 in the second year, £3 in the third year, and so on...*
- *...With payments made at the start of each year,...*
- *...As long as the life is alive.*



Amount	£1	£2	£3	£4	···
Discount	1	v	v^2	v^3	···
Probability	1	p_x	${}_2p_x$	${}_3p_x$	···

The present value random variable of these payments is

$$Y := \sum_{k=0}^{\infty} (k+1)\, \mathbb{1}_{[K_x \geq k]}\, v^k = (I\ddot{a})_{\overline{K_x+1}|}.$$

Recall that for an n-year increasing annuity-certain,

$$(I\ddot{a})_{\overline{n}|} = \frac{\ddot{a}_{\overline{n}|} - n\, v^n}{d}.$$

The expected value of Y is

$$(I\ddot{a})_x := \mathbb{E}\,(Y) = \mathbb{E}\left((I\ddot{a})_{\overline{K_x+1}|}\right) = \mathbb{E}\left(\frac{\ddot{a}_{\overline{K_x+1}|} - (K_x+1)\, v^{K_x+1}}{d}\right) = \frac{\ddot{a}_x - (IA)_x}{d},$$

i.e.

$$(I\ddot{a})_x = \frac{\ddot{a}_x - (IA)_x}{d}.$$

Example 5.27 *Consider an arithmetically increasing term annuity-due in which:*

- *A payment of £1 is made in the first year, a payment of £2 in the second year, £3 in the third year, and so on...*
- *...With payments made at the start of each year,...*
- *...As long as the life is alive and...*
- *...For a maximum of an integer $n > 0$ years.*

Following a similar argument to the arithmetically increasing whole life annuity-due, we can show that the EPV of the present value of an arithmetically increasing n-year term annuity-due is

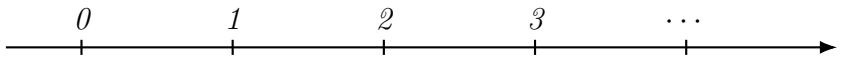
$$(I\ddot{a})_{x:\overline{n}|} = \frac{\ddot{a}_{x:\overline{n}|} - (IA)_{x:\overline{n}|}}{d}.$$

5.10 Geometrically increasing annuities

Many life annuity contracts have annual increases to the annuity payments, for example, the payments increase in line with a price inflation index or at a fixed rate such as 3% per annum. While such increases are very common, there is no international actuarial notation for them. Instead, the interest rate used for discounting the payments is adjusted to include the increases.

Example 5.28 *Consider a geometrically increasing whole life annuity-due in which:*

- *A payment of £1 is made in the first year, a payment of £$(1+g)$ in the second year, £$(1+g)^2$ in the third year, and so on...*
- *...With payments made at the start of each year,...*
- *...As long as the life is alive.*



Amount	£1	£$(1+g)^1$	£$(1+g)^2$	£$(1+g)^3$	$\cdots$
Discount	1	v	v^2	v^3	$\cdots$
Probability	1	p_x	${}_2p_x$	${}_3p_x$	$\cdots$

The present value random variable of these payments is

$$Y := \sum_{k=0}^{\infty} (1+g)^k \, \mathbb{1}_{[K_x \geq k]} \, v^k = \sum_{k=0}^{\infty} \left(\frac{1+g}{1+i} \right)^k \mathbb{1}_{[K_x \geq k]}.$$

Define

$$i^\star := \frac{1+i}{1+g} - 1.$$

Then

$$\mathbb{E}(Y) = \mathbb{E}\left(\sum_{k=0}^{\infty} \left(\frac{1}{1+i^\star} \right)^k \mathbb{1}_{[K_x \geq k]} \right) = \sum_{k=0}^{\infty} \left(\frac{1}{1+i^\star} \right)^k \mathbb{P}[K_x \geq k] = \sum_{k=0}^{\infty} \left(\frac{1}{1+i^\star} \right)^k {}_kp_x = \ddot{a}_x,$$

where $\ddot{a}_x$ is the EPV of a whole life annuity-due paying £1 per annum and valued at the effective rate of interest $i^\star$ p.a.

5.11 Annuities payable m times per year

Premiums paid by the policyholders are usually paid monthly, quarterly or even weekly. Similarly, annuity payments to a retiree are usually paid monthly or weekly.

Fix a positive integer $m > 0$. Recall that the random variable $K_x^{(m)}$ is the future lifetime of (x), expressed in years, rounded to the lower $1/m$th of a year, i.e.

$$K_x^{(m)} := \frac{1}{m} \lfloor mT_x \rfloor,$$

where $\lfloor mT_x \rfloor$ is the integer part of mT_x.

The probability mass function of $K_x^{(m)}$ is for $k = 0, \frac{1}{m}, \frac{2}{m}, \ldots,$

$$\mathbb{P}\left[K_x^{(m)} = \frac{k}{m}\right] = \mathbb{P}\left[\frac{k}{m} \leq T_x < \frac{k+1}{m}\right] = {}_{\frac{k}{m}|\frac{1}{m}}q_x = {}_{\frac{k}{m}}p_x - {}_{\frac{k+1}{m}}p_x.$$

Note that the probability that (x) lives at least $\frac{k}{m}$ years is

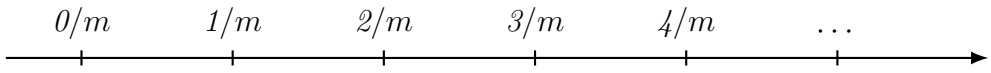
$$\mathbb{P}\left[K_x^{(m)} \geq \frac{k}{m}\right] = \sum_{r=k}^{\infty} \mathbb{P}\left[K_x^{(m)} = \frac{r}{m}\right] = \sum_{r=k}^{\infty} \left({}_{\frac{r}{m}}p_x - {}_{\frac{r+1}{m}}p_x\right) = {}_{\frac{k}{m}}p_x.$$

It can be easier to set $m = 12$ and read '$m$thly' as 'monthly', when reading the text in this section.

5.11.1 Whole life annuity-due payable $m$thly

Example 5.29 *To value a* whole life annuity-due payable $m$thly*, consider an annuity where:*

- *Payments of £$\frac{1}{m}$ are made m times a year, at the start of each $1/m$ year,...*
- *...As long as a life, initially aged x years, is alive.*



Amount	£$\frac{1}{m}$	£$\frac{1}{m}$	£$\frac{1}{m}$	£$\frac{1}{m}$	£$\frac{1}{m}$	...
Discount	1	$v^{1/m}$	$v^{2/m}$	$v^{3/m}$	$v^{4/m}$	...
Probability	1	${}_{\frac{1}{m}}p_x$	${}_{\frac{2}{m}}p_x$	${}_{\frac{3}{m}}p_x$	${}_{\frac{4}{m}}p_x$	...

Let the effective rate of interest be $i > -1$ p.a. The present value random variable of these payments is

$$Y := \frac{1}{m} \sum_{k=0}^{\infty} \mathbb{1}_{[K_x^{(m)} \geq \frac{k}{m}]}\, v^{\frac{k}{m}}.$$

The expected value of Y *is* $\ddot{a}_x^{(m)} := \mathbb{E}(Y)$. *As* $\mathbb{P}\left[K_x^{(m)} \geq \frac{k}{m}\right] = {}_{\frac{k}{m}}p_x$,

$$\mathbb{E}(Y) = \frac{1}{m}\,\mathbb{E}\left(\sum_{k=0}^{\infty} \mathbb{1}_{[K_x^{(m)} \geq \frac{k}{m}]}\, v^{\frac{k}{m}}\right) = \frac{1}{m}\sum_{k=0}^{\infty} \mathbb{P}\left[K_x^{(m)} \geq \frac{k}{m}\right] v^{\frac{k}{m}} = \frac{1}{m}\sum_{k=0}^{\infty} {}_{\frac{k}{m}}p_x\, v^{\frac{k}{m}}.$$

Hence

$$\boxed{\ddot{a}_x^{(m)} = \frac{1}{m}\sum_{k=0}^{\infty} {}_{\frac{k}{m}}p_x\, v^{\frac{k}{m}}.}$$

Notice that the amount $£\frac{1}{m}$ *has 'disappeared' in* $\ddot{a}_x^{(m)}$. *The same happens for the notation* $a_{\overline{n}|}^{(m)}$; *the total annual payment is* $£1$ *but the superscript* $^{(m)}$ *means that the payment is* $£\frac{1}{m}$ *every* $\frac{1}{m}$ *year.*

Exercise 5.30 *Explain why the present value random variable of the whole life annuity-due payable mthly can be expressed as*

$$Y = \ddot{a}_{\overline{K_x^{(m)}+\frac{1}{m}|}}^{(m)}.$$

For this annuity, there are $m\,K_x^{(m)} + 1$ payments in total, lasting for $K_x^{(m)} + \frac{1}{m}$ years (where $\frac{1}{m}$ year is added as the payments are at the start of each $m$thly period).

	$0/m$	$1/m$	$2/m$	$\cdots$	$K_x^{(m)}$	$K_x^{(m)} + \frac{1}{m}$	Time
Amount	$£\frac{1}{m}$	$£\frac{1}{m}$	$£\frac{1}{m}$	$\cdots$	$£\frac{1}{m}$		
Discount	1	$v^{1/m}$	$v^{2/m}$	$\cdots$	$v^{K_x^{(m)}}$		
Probability	1	${}_{\frac{1}{m}}p_x$	${}_{\frac{2}{m}}p_x$	$\cdots$	${}_{K_x^{(m)}}p_x$		

Exercise 5.31 *Show that*

$$\boxed{\ddot{a}_x^{(m)} = \frac{1 - A_x^{(m)}}{d^{(m)}}.}$$

From the last expression for the present value,

$$Y = \ddot{a}_{\overline{K_x^{(m)}+\frac{1}{m}|}}^{(m)} = \frac{1 - v^{K_x^{(m)}+\frac{1}{m}}}{d^{(m)}}.$$

Thus

$$\ddot{a}_x^{(m)} = \mathbb{E}(Y) = \mathbb{E}\left(\frac{1 - v^{K_x^{(m)}+\frac{1}{m}}}{d^{(m)}}\right) = \frac{1 - \mathbb{E}(v^{K_x^{(m)}+\frac{1}{m}})}{d^{(m)}}.$$

Since $v^{K_x^{(m)}+\frac{1}{m}}$ represents a payment of 1 unit paid at the end of the $\frac{1}{m}$ year of death of (x), then $\mathbb{E}(v^{K_x^{(m)}+\frac{1}{m}}) = A_x^{(m)}$. Substituting into the last fraction, we get the result.

5.11.2 Term annuity-due payable $m$thly

Example 5.32 *To value a* term annuity-due payable $m$thly, *consider an annuity where:*

- *Payments of* $£\frac{1}{m}$ *are made* m *times a year, at the start of each* $1/m$ *year,...*
- *...As long as a life, initially aged* x *years, is alive...*
- *...For a maximum of an integer* $n > 0$ *years.*

The expected present value of these payments is

$$\boxed{\ddot{a}^{(m)}_{x:\overline{n}|} = \frac{1}{m} \sum_{k=0}^{nm-1} {}_{\frac{k}{m}}p_x \, v^{\frac{k}{m}}.}$$

Exercise 5.33 *Show that*

$$\boxed{\ddot{a}^{(m)}_{x:\overline{n}|} = \ddot{a}^{(m)}_{x} - v^n \, {}_np_x \, \ddot{a}^{(m)}_{x+n}.}$$

$$\begin{aligned}
\ddot{a}^{(m)}_{x:\overline{n}|} = \frac{1}{m} \sum_{k=0}^{nm-1} {}_{\frac{k}{m}}p_x \, v^{\frac{k}{m}} &= \frac{1}{m} \sum_{k=0}^{\infty} {}_{\frac{k}{m}}p_x \, v^{\frac{k}{m}} - \frac{1}{m} \sum_{k=nm}^{\infty} {}_{\frac{k}{m}}p_x \, v^{\frac{k}{m}} \\
&= \ddot{a}^{(m)}_{x} - \frac{1}{m} \sum_{r=0}^{\infty} {}_{\frac{r+nm}{m}}p_x \, v^{\frac{r+nm}{m}} \\
&= \ddot{a}^{(m)}_{x} - {}_np_x \, v^n \, \frac{1}{m} \sum_{r=0}^{\infty} {}_{\frac{r}{m}}p_{x+n} \, v^{\frac{r}{m}} \\
&= \ddot{a}^{(m)}_{x} - {}_np_x \, v^n \, \ddot{a}^{(m)}_{x+n}.
\end{aligned}$$

5.11.3 Other annuities payable $m$thly

It should be clear that, for example, $a^{(m)}_x$ represents a whole life immediate annuity payable $m$thly and $a^{(m)}_{x:\overline{n}|}$ represents an n-year term immediate annuity payable $m$thly.

Exercise 5.34 *Show that*

$$a^{(m)}_x = \ddot{a}^{(m)}_x - \frac{1}{m}$$

and

$$a^{(m)}_{x:\overline{n}|} = \ddot{a}^{(m)}_{x:\overline{n}|} - \frac{1}{m} + \frac{1}{m} \, {}_np_x \, v^n.$$

As the first possible payment for a whole life immediate annuity payable $m$thly is at time $\frac{1}{m}$,

$$a_x^{(m)} = \frac{1}{m}\sum_{k=1}^{\infty} {}_{\frac{k}{m}}p_x\, v^{\frac{k}{m}} = \frac{1}{m}\sum_{k=0}^{\infty} {}_{\frac{k}{m}}p_x\, v^{\frac{k}{m}} - \frac{1}{m}\,{}_{\frac{0}{m}}p_x\, v^{\frac{0}{m}} = \ddot{a}_x^{(m)} - \frac{1}{m}.$$

For the second result, the first possible payment for an n-year term immediate annuity payable $m$thly is at time $\frac{1}{m}$ and the last possible payment is at time n

$$a_{x:\overline{n}|}^{(m)} = \frac{1}{m}\sum_{k=1}^{nm} {}_{\frac{k}{m}}p_x\, v^{\frac{k}{m}} = \frac{1}{m}\sum_{k=0}^{nm-1} {}_{\frac{k}{m}}p_x\, v^{\frac{k}{m}} - \frac{1}{m}\,{}_{\frac{0}{m}}p_x\, v^{\frac{0}{m}} + \frac{1}{m}\,{}_{\frac{nm}{m}}p_x\, v^{\frac{nm}{m}} = \ddot{a}_{x:\overline{n}|}^{(m)} - \frac{1}{m} + \frac{1}{m}\,{}_np_x\, v^n.$$

5.12 Recursions

Suppose we have the one-year survival probabilities $\{p_x; x = 0, 1, 2, \ldots, \omega - 1\}$, where the integer $\omega < \infty$ is the limiting age of the population. To calculate the EPVs of an annuity at these ages, we can use a recursion formula.

Exercise 5.35 *Show that*

$$\boxed{\ddot{a}_x = 1 + v\, p_x\, \ddot{a}_{x+1}.}$$

$$\begin{aligned}\ddot{a}_x = \sum_{k=0}^{\infty} v^k\, {}_kp_x = 1 + \sum_{k=1}^{\infty} v^k\, {}_kp_x = 1 + \sum_{r=0}^{\infty} v^{r+1}\, {}_{r+1}p_x &= 1 + v\, p_x \sum_{r=0}^{\infty} v^r\, {}_rp_{x+1} \\ &= 1 + v\, p_x\, \ddot{a}_{x+1}.\end{aligned}$$

Example 5.36 *Suppose the limiting age of a population is $\omega = 114$ years. Given the one-year survival probabilities shown in the table below, the EPVs of a whole life annuity-due paying £1 per annum can be calculated using the recursion for a chosen interest rate. Here, choose $i = 0.05$ per annum.*

A starting value must be known for the recursion. Here, we must have $\ddot{a}_{114} = 0$ since everyone is dead by the limiting age 114 in this survival model.

x	p_x	$1 + (1.05)^{-1}\, p_x\, \ddot{a}_{x+1} =$	$\ddot{a}_x$
114	0		0
113	0	$1 + (1.05)^{-1}\,(0)\,(0) =$	1
112	0.39745	$1 + (1.05)^{-1}\,(0.39745)\,(1) =$	1.3785
111	0.42068	$1 + (1.05)^{-1}\,(0.42068)\,(1.3785) =$	1.5523
110	0.44426	$1 + (1.05)^{-1}\,(0.44426)\,(1.5523) =$	1.6568
⋮	⋮	⋮	⋮

Exercise 5.37 *Show that*

$$\boxed{\ddot{a}_x^{(m)} = \frac{1}{m} + v^{\frac{1}{m}}\, {}_{\frac{1}{m}}p_x\, \ddot{a}_{x+\frac{1}{m}}^{(m)}\,.}$$

$$\begin{aligned}
\ddot{a}_x^{(m)} = \frac{1}{m}\sum_{k=0}^{\infty} {}_{\frac{k}{m}}p_x\, v^{\frac{k}{m}} = \frac{1}{m} + \frac{1}{m}\sum_{k=1}^{\infty} {}_{\frac{k}{m}}p_x\, v^{\frac{k}{m}} &= \frac{1}{m} + \frac{1}{m}\sum_{r=0}^{\infty} {}_{\frac{r+1}{m}}p_x\, v^{\frac{r+1}{m}} \\
&= \frac{1}{m} + v^{\frac{1}{m}}\, {}_{\frac{1}{m}}p_x\, \frac{1}{m}\sum_{r=0}^{\infty} {}_{\frac{r}{m}}p_{x+\frac{1}{m}}\, v^{\frac{r}{m}} \\
&= \frac{1}{m} + v^{\frac{1}{m}}\, {}_{\frac{1}{m}}p_x\, \ddot{a}_{x+\frac{1}{m}}^{(m)}\,.
\end{aligned}$$

Example 5.38 *Suppose the limiting age of a population is $\omega = 100$ years. Given the one-year survival probabilities shown in the table below, the EPVs of a whole life annuity-due paying £1 per annum monthly can be calculated using the recursion for a chosen interest rate. Here, choose $i^{(12)} = 0.06$ per annum. In that case,*

$$1 + i = \left(1 + \frac{i^{(12)}}{12}\right)^{12} = \left(1 + \frac{0.06}{12}\right)^{12} = (1.005)^{12}\,,$$

so

$$v^{\frac{1}{12}} = (1+i)^{-\frac{1}{12}} = (1.005)^{-1}\,.$$

As before, set $\ddot{a}_{100} = 0$ as the starting value, since everyone is dead by the limiting age 100.

x	${}_{\frac{1}{12}}p_x$	$\frac{1}{12} + v^{\frac{1}{12}}\, {}_{\frac{1}{12}}p_x\, \ddot{a}_{x+\frac{1}{12}}^{(12)} =$	$\ddot{a}_x^{(12)}$
100	0		0
$99\frac{11}{12}$	0	$\frac{1}{12} + (1.005)^{-1}\,(0)\,(0) =$	$\frac{1}{12}$
$99\frac{10}{12}$	0.955	$\frac{1}{12} + (1.005)^{-1}\,(0.955)\,(\frac{1}{12}) =$	0.1625
$99\frac{9}{12}$	0.947	$\frac{1}{12} + (1.005)^{-1}\,(0.947)\,(0.1625) =$	0.2365
$99\frac{8}{12}$	0.921	$\frac{1}{12} + (1.005)^{-1}\,(0.921)\,(0.2365) =$	0.3000
⋮	⋮	⋮	⋮

5.13 Evaluating EPVs of annuities using Woolhouse's formula

What if we want to calculate the EPV of a monthly paid annuity, but we have only the EPVs of annuities paid annually?

The survival probabilities are often given only for integer ages, but the EPV of a monthly paid annuity requires knowledge of survival between integer ages. So we cannot go straight back to basics, and calculate the EPV of a monthly paid annuity from survival probabilities and the interest rate using the appropriate recursion formula alone. We must make further assumptions.

We present a method of calculating the EPV of an $m$thly paid annuity using the EPVs of annuities paid annually, e.g. to calculate $\ddot{a}_{60}^{(12)}$ using the value $\ddot{a}_{60}$. The method is called *Woolhouse's formula*. It is based on the Euler-Maclaurin formula for numerical integration.

Read Dickson, Hardy & Waters, Section 5.11.3 for the derivation of Woolhouse's formula. The derivation is not examinable.

We use only the first-order terms as Woolhouse's formula; you will see in Dickson, Hardy & Waters, Section 5.11.3, equation (5.43), that there is a second-order term. We do not use the second-order term in this course.

Definition 5.39 *Woolhouse's formula for the EPV of an mthly-paid whole life annuity-due is*

$$\ddot{a}_x^{(m)} \approx \ddot{a}_x - \frac{m-1}{2m}.$$

Letting $m \to \infty$, Woolhouse's formula for the EPV of a whole life continuous annuity is

$$\bar{a}_x^{(m)} \approx \ddot{a}_x - \frac{1}{2}.$$

Exercise 5.40 *By writing $\ddot{a}_{x:\overline{n}|}^{(m)}$ as the difference of two whole life annuities, show that*

$$\ddot{a}_{x:\overline{n}|}^{(m)} \approx \ddot{a}_{x:\overline{n}|} - \frac{m-1}{2m}\left(1 - v^n\, {}_np_x\right).$$

From Exercise 5.33,

$$\ddot{a}_{x:\overline{n}|}^{(m)} = \ddot{a}_x^{(m)} - v^n\, {}_np_x\, \ddot{a}_{x+n}^{(m)}.$$

Apply Woolhouse's formula to $\ddot{a}_x^{(m)}$ and $\ddot{a}_{x+n}^{(m)}$ to get

$$\ddot{a}_{x:\overline{n}|}^{(m)} \approx \left(\ddot{a}_x - \frac{m-1}{2m}\right) - v^n\, {}_np_x \left(\ddot{a}_{x+n} - \frac{m-1}{2m}\right)$$

As $\ddot{a}_{x:\overline{n}|} = \ddot{a}_x - v^n\,{}_np_x\,\ddot{a}_{x+n}$,

$$\ddot{a}^{(m)}_{x:\overline{n}|} \approx \ddot{a}_{x:\overline{n}|} - \frac{m-1}{2m} + v^n\,{}_np_x\,\frac{m-1}{2m} = \ddot{a}_{x:\overline{n}|} - \frac{m-1}{2m}\left(1 - v^n\,{}_np_x\right).$$

Exercise 5.41 *By writing $\bar{a}^{(m)}_{x:\overline{n}|}$ as the difference of two whole life annuities, show that*

$$\bar{a}_{x:\overline{n}|} \approx \ddot{a}_{x:\overline{n}|} - \frac{1}{2}\left(1 - v^n\,{}_np_x\right).$$

From Exercise 5.22,

$$\bar{a}_{x:\overline{n}|} = \bar{a}_x - v^n\,{}_np_x\,\bar{a}_{x+n}.$$

Apply Woolhouse's formula to $\bar{a}_x$ and $\bar{a}_{x+n}$ to get

$$\bar{a}_{x:\overline{n}|} \approx \left(\ddot{a}_x - \frac{1}{2}\right) - v^n\,{}_np_x\,\left(\ddot{a}_{x+n} - \frac{1}{2}\right)$$

As $\ddot{a}_{x:\overline{n}|} = \ddot{a}_x - v^n\,{}_np_x\,\ddot{a}_{x+n}$,

$$\bar{a}_{x:\overline{n}|} \approx \ddot{a}_{x:\overline{n}|} - \frac{1}{2} + v^n\,{}_np_x\,\frac{1}{2} = \ddot{a}_{x:\overline{n}|} - \frac{1}{2}\left(1 - v^n\,{}_np_x\right).$$

Handy hint 5.42 *You can find the formulae*

$$\ddot{a}^{(m)}_x \approx \ddot{a}_x - \frac{m-1}{2m} \quad \textit{and} \quad \ddot{a}^{(m)}_{x:\overline{n}|} \approx \ddot{a}_{x:\overline{n}|} - \frac{m-1}{2m}\left(1 - v^n\,{}_np_x\right)$$

in Formulae and Tables for Actuarial Examinations*, in the* Formulae *section, Section 6.*

Remark 5.43 *Another method of calculating the EPV of an mthly paid annuity using the EPVs of annuities paid annually, uses the Uniform Distribution of Deaths (UDD) assumption between integer ages. While you can read about the method in Dickson, Hardy & Waters, Section 5.11.2, this course no longer includes it. Instead, only Woolhouse's formula will be used.*